

Performance Testing

Field	Details
Starting Date	08 October 2026
Team ID	UNM1611Firefox
Project Name	AI STUDY BUDDY API

Step 1: Testing Overview

Purpose: assess the API's responsiveness and stability for its implemented authentication, study-material, AI-generation, and admin operations. The supplied project does not contain an automated performance-test suite, so this document records a reproducible performance-test plan and clearly separates planned measurements from results that were not evidenced in the supplied ZIP.

Area	Implemented feature / endpoint group	Performance focus	Status
Authentication	POST /register, /login, /refresh, /logout	Response time, concurrent login/refresh handling	Planned measurement
Materials	POST /upload, GET /, GET /:id, DELETE /:id	Upload latency, list/read latency, 5 MB boundary	Planned measurement
AI generation	POST /:id/summarize, /flashcards, /quiz, /study-plan	Gemini-dependent latency and timeout/error behaviour	Planned measurement
Admin	GET /users, DELETE /users/:id, GET /stats	Read/delete response time under multiple requests	Planned measurement

Marks / Evaluation Allocation

The original GitHub template contains evaluation/marks areas. Because the repository connector exposes the PDF as binary but does not expose its rendered mark values for reliable extraction, no mark value has been invented here. The marks/evaluation field is intentionally retained for evaluator entry.

Evaluation / Marks	Project evidence	Score / Marks
Performance test coverage	Authentication, materials, AI, admin scenarios documented	Evaluator entry
Performance results	No executed performance log supplied with project ZIP	Evaluator entry
Analysis & observations	Risks, bottlenecks and recommendations recorded	Evaluator entry

Step 2: Test Scenarios

ID	Scenario	Input / Load	Expected observation	Result
PT-01	Register users	10, 25, 50 sequential/concurrent requests	Stable 2xx responses for valid unique users; duplicate email rejected	Not executed
PT-02	Login	10, 25, 50 requests	Consistent authentication latency; no unexpected 5xx errors	Not executed
PT-03	Refresh token	Repeated valid refresh requests	Stable token refresh latency and correct error handling for invalid tokens	Not executed
PT-04	Material list	Repeated GET /materials	Latency remains stable as material count grows	Not executed
PT-05	Material upload	Small text/markdown/PDF plus near-5 MB file	Upload completes within acceptable threshold; >5 MB rejected	Not executed
PT-06	AI summary	Repeated summary generation	Latency dominated by Gemini call; errors handled without server crash	Not executed
PT-07	AI flashcards/quiz	Repeated generation requests	Valid JSON response parsing; service remains responsive	Not executed
PT-08	Study plan	Repeated generation requests	Stable response and controlled failure behaviour	Not executed
PT-09	Admin operations	Repeated users/stats requests	Role-protected endpoints remain responsive	Not executed
PT-10	Mixed workload	Concurrent auth + material + AI calls	No process crash; identify bottleneck category	Not executed

Suggested load levels: 10, 25 and 50 concurrent requests for API endpoints; AI endpoints should be tested separately because external Gemini latency can dominate the measurement.

Step 3: Performance Test Results

Metric	Target / measurement rule	Observed result	Evidence
Average response time	Measure per endpoint under each load level	Not measured	No performance log in ZIP
P95 response time	Record 95th percentile	Not measured	No performance log in ZIP
Requests / second	Calculate successful requests per second	Not measured	No performance log in ZIP
Error rate	5xx + failed requests / total requests	Not measured	No performance log in ZIP
Upload performance	Measure by file size, especially near 5 MB	Not measured	No performance log in ZIP
AI latency	Measure each Gemini-backed operation separately	Not measured	External dependency; no run log

Step 4: Observations & Analysis

1. AI generation is expected to be the dominant latency source because summarization, flashcards, quiz generation and study-plan generation call Google Gemini.
2. File upload performance is affected by Multer processing and synchronous file reading in the material controller.
3. MongoDB operations are used for users and study materials; database indexing and connection behaviour should be monitored as data volume increases.
4. The project has a 5 MB upload limit, which gives a clear upper boundary for upload tests.
5. The supplied package has no real automated test suite; package.json contains a placeholder test script. Therefore no fabricated pass/fail or latency numbers are reported.
6. AI JSON outputs are parsed with JSON.parse without an additional schema-validation layer; malformed model output should be included in resilience testing.

Step 5: Screenshots / Evidence

No performance-test screenshots or execution logs were included in the supplied project ZIP. This section is intentionally marked as evidence pending rather than presenting invented measurements.

